

Lab 5 CS2 – Experiment with Skip List and Tree Set

In this week's lab you will need to perform the following tasks:

1. Add a delete function to the provided Skip List code that we have written in the lecture. Make sure to test your delete code to see whether it can delete an item properly from the list.
2. Perform experiment as discussed below and plot the experimental result in an excel file
3. Upload the Main.java and the excel file in webcourses

You will perform some experiments with skip list and tree set to see how they scale with the number of data and compare them.

Write a function that takes the test size as parameter and then based on that your code will randomly generate two set of test size number of integers within the range of 0 to 2000000. The first set of numbers will be used for insertion and the second set of numbers will be used for deletion.

Before the insertion process, start the timer using `"System.currentTimeMillis();"`

Then insert all the items to your skip list

Then delete the list of items you have generated from the skip list

Stop the timer and calculate the time taken

Do, the same thing for Treemap.

In the main function, declare this array and pass each of them to the test functions

Run your code 3 times and enter those data in the excel file in a well formatted way. Plot the average. There should be three plots, one for insertion time, one for deletion time, and one for total time. For each figure, it should contain the plot for both skip list and tree set so that they can be compared from the same figure

The output of your code should be something like this:

Test size: 50000

=====

Skip list insertion took 10 ms.
Skip list deletion took 10 ms.
Skip list actions took 10 ms.
tree set insertion took 10 ms.
tree set deletion took 10 ms.
tree set actions took 10 ms.

Test size: 100000

=====

Skip list insertion took 10 ms.
Skip list deletion took 10 ms.
Skip list actions took 10 ms.
tree set insertion took 10 ms.
tree set deletion took 10 ms.
tree set actions took 10 ms.

Test size: 150000

=====

Skip list insertion took 10 ms.
Skip list deletion took 10 ms.
Skip list actions took 10 ms.
tree set insertion took 10 ms.
tree set deletion took 10 ms.
tree set actions took 10 ms.

Test size: 200000

=====

Skip list insertion took 10 ms.
Skip list deletion took 10 ms.
Skip list actions took 10 ms.
tree set insertion took 10 ms.
tree set deletion took 10 ms.
tree set actions took 10 ms.

Test size: 250000

=====

Skip list insertion took 10 ms.
Skip list deletion took 10 ms.
Skip list actions took 10 ms.
tree set insertion took 10 ms.
tree set deletion took 10 ms.
tree set actions took 10 ms.

Test size: 300000

=====

Skip list insertion took 10 ms.
Skip list deletion took 10 ms.
Skip list actions took 10 ms.
tree set insertion took 10 ms.
tree set deletion took 10 ms.
tree set actions took 10 ms.

Test size: 350000

=====

! upto 500k